Three stages of temporal interference

Draft21: whole-paper legibility revision approved on 20 October 2026

Temporal-interference (TI) stimulation requires three stages in series: the kilohertz field must reach a nonlinear membrane (access), the membrane must generate the difference frequency absent from the field (detection), and the circuit must amplify that weak component (amplification). We build a mesoscale model. Following single-neuron TI models, we represent access by a short effective membrane time. We analyze detection and amplification in a Jansen–Rit column, an exact mean field of quadratic integrate-and-fire neurons, and its spiking network. For weak fields the detector is quadratic: the beat is proportional to the product of the local carrier amplitudes, and a signed steady source to their summed squares. A static rate curve and the exact mean field give the same product law; the mean field adds frequency dependence when membranes are slow relative to input. In a realistic head with a published hippocampal montage, changing the current ratio moves the envelope and mean-square foci in opposite directions and only rescales the product map. Near a Hopf bifurcation the circuit amplifies the beat as the inverse of the modal damping; with the Jansen–Rit operating point held fixed, synaptic coupling sets this gain independently of detection. The amplified response appears mainly in envelope-locked timing, with little change in mean rate. At stronger fields the steady shift alone can drive a gamma circuit across its oscillation threshold, as can an unmodulated carrier of equal power. The model expresses TI efficacy as the product of three separately measurable factors. For weak fields, dose follows the product of carrier amplitudes rather than the envelope minimum. Under uniform conductivity, access, and detection, the steady drive has its largest magnitude at compartment boundaries. TI shares its circuit amplifier with direct low-frequency stimulation. Carrier sweeps, independent-amplitude tests, beat sweeps across states, and equal-power controls each test one stage.

The shared abstract retains its approved claims. This pass changes explanation and organization, not simulations or numerical evidence. Draft20 (`e20c047`) is the comparison baseline; Giulio's annotated PDF is preserved. The revision is built separately under `build/review/`. Pending Ricardo orientation and interface results are outside this prose pass.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Abstract and Introduction	A two-tone field has no difference-frequency line. Access creates local polarization; detection generates a beat and steady source; amplification determines the response. Retain the cellular and network literature, weak-perturbation scope, and approved FEM abstract sentence.	The reader learns the physical problem before the model equations. The tests distinguish the three stages rather than assigning every response to demodulation.
Theory: population curve and field coupling	Keep the JR sigmoid, synaptic response, and direct polarization. Derive the quadratic cross term and constant term together. Define and number the dynamic coefficient; retain its weak-input, memory, and nonzero-coefficient conditions.	The rate curve first explains what mixes the carriers. The NMM1 operator notation moves to the model appendix; it is not needed to understand this step.
Theory: envelope, spatial weighting, harmonics	Define the spatially varying envelope before using its harmonics. Retain the side-by-side map table, product/minimum relation, two-site example, fixed-pair current-ratio law, signed projections, and recurrent weighting. Retain the harmonic ladder and mixed-detector equation.	The reader can distinguish an envelope map from a detector output, then see how amplitude and harmonic measurements separate their predictions.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Theory: steady source	Retain the distinction between nonnegative mean-square polarization, signed source, and circuit response. Retain the convex/concave sign, polarity invariance, beat/DC inequality, 202-fold example, both subharmonicity arguments, and their uniformity conditions.	The spatial restriction concerns the mean map under stated conditions; it does not prohibit an interior beat maximum. Orientation and equal-power qualifications stay here.
Theory: gain, access, operating point	Retain the synaptic attenuation estimates, resonant susceptibility, access–detection–gain product, static versus dynamic coefficients, access-time table, cable estimate, measured versus assumed times, and PEIX identities.	After detection is understood, the reader sees what suppresses the carriers and amplifies the beat, and why this does not establish physiological access.
Theory: exact mean field	Move the QIF introduction after the static argument. Keep the MPR equations in physical time and rate, its nonlinear equilibrium curve, threshold recruitment, and slow-synapse and carrier-relaxation conditions. Retain the signs and asymptotes of dynamic mixing.	QIF now answers a question already posed: when does a static curve suffice? The exact rational coefficient, fixed-voltage coefficient, and carrier-exponent table move intact to the supplement.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Methods	State the purpose of waveform matching, lock-in, FEM geometry, and each model comparison before settings. Put projection choices with the head model. Place the shortened circuit schematic beside model equations. Retain all run parameters and finite-network protocols.	Methods connects each test to its observable. Detailed JR run settings, PING endpoint stability, and QIF cutoff results have explicit supplementary homes.
Results: JR and FEM	Combine the two-sentence beat demonstration with resonance. Retain the two-tone spectrum, curvature, square-law and linearization controls, kHz access checks, FEM maps and sensitivities, pinned coupling sweep, timing ratios, and TI/tACS controls.	The evidence proceeds from beat generation to spatial consequences and recurrent amplification. Captions identify panels and conditions; interpretation belongs in the text.
Results: NMM2 and spikes	Retain dynamic-source accuracy, the two-mode coupling curve, gain exponent and finite-amplitude deviations, attracting endpoint, spike cutoff qualification, both field amplitudes, equal-RMS carrier control, sidebands, and phase statistics.	The exact model confirms mixing and gain, then the network separates beat modulation from induced gamma. Numerical diagnostics are consolidated rather than repeated in captions.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Discussion	Group interpretation into access/mixing, spatial predictions, field-induced oscillations and gain, efficiency, population-model limits, and tests. Retain all Violante statistics and physiological conditions. Preserve the Vieira comparison and Cifra scope in attached footnotes.	Each subsection answers one scientific question. The efficiency argument distinguishes generated drive from PLV and keeps the numerical ceiling within its stated family.
Predictions and Conclusion	Retain every test row, independent amplitudes, fixed-pair steering, state dependence, harmonic calibration, timing, blockade, and equal-power controls. Put the off-resonance measurement design beside these tests.	The conclusion states the supported findings and the measurements that separate mechanisms; detailed qualifications remain beside their claims.
Notation and supplement	Replace tilde variables with explicit physical units in the article and an explicit membrane-time convention in the uncoupled derivation. Keep all equations, citations, supplementary figures, LaNMM results, detector-bound derivation, and vocabulary metrics.	The supplement contains the derivations and diagnostics needed to reproduce the argument. It preserves all material relocated from the article.

Changes and review conditions

No substantive scientific removal is proposed in this pass

The NMM1 operator formulation moves from the opening to the model appendix. The QIF equilibrium and slow-synapse argument move to the final Theory subsection; the full mixing coefficient and exponent table move to the supplement. Anatomical projection choices move to the FEM method. The model schematic moves beside the model equations, with pathway definitions retained there and in the appendix. The harmonic-test design moves from Methods to the predictions discussion. Detailed JR run settings, PING endpoint stability, and finite-cutoff diagnostics move to dedicated supplementary subsections. Figure interpretations and conditions remain in the neighboring text when their captions are shortened.

Repeated statements are consolidated, not used to justify deleting conditions, examples, or results. The earlier seven CUT? items remain pending and retained. The title, authorship, simulation caches, and approved numerical results are unchanged. Public GitHub, Calliope, Zenodo, and coauthor circulation remain held.

Review requires a continuous read of the rendered article and supplement, followed by checks of retained labels and citations, notation, cross-references, and builds. Klaus supplies an independent reader review of the frozen revision. A clean build and a fidelity check alone do not establish readability.